

CORRELATION

1D $len(w) = 5$

S

W

zero-padding

1) 0000010000000
12428

$L > (1 \cdot 0) + (2 \cdot 0) + (4 \cdot 0) + (2 \cdot 0) + (8 \cdot 0) = 0$

2) 0000010000000
12428

$L \rightarrow 0 \cdot 0 + 0 \cdot 0 + 8 = 8$

3) 0000010000000
12428

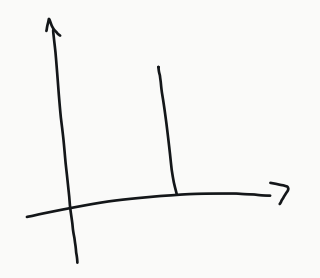
$= 2$

4) 0000010000000
12428

$= 4$

5) 0000010000000
12428

$= 2$



- 6) 1
- 7) 4
- 8) 4
- 08242100 "Same" $L_s = L_f$
- 000824210000 "Full" $L_s < L_f$
- 2421 "Valid" $L_s > L_f$

CONVOLUTION

S

W

1) 0000010000000
82421

L_s

0000010000000
82421

$= 0$

0000010000000
82421

$= 1$

0000010000000
82421

$= 2$

0000010000000
82421

$= 4$

0000010000000
82421

$= 2$

0000010000000
82421

$= 8$

0000010000000
82421

$= 0$

0000010000000
82421

$= 0$

$W^{avg} = \frac{1}{2, 4, 2, 8}$

$\rightarrow W^{avg} = 0.2421$

01242800

$$1) \sum(0, 40, 0, 13, 10, 0, 5, 7) = 20 \quad \bigg| \quad 2) \sum(13, 10, 10, 11, 5, 7) = 130$$

3x3

200

13	10	100	
5	7	9	
6	6	1	

= 40k

